

Report of Findings

Findings

The Exploratory Data Analysis (EDA) of the Supermart Grocery Sales dataset provides several important insights into the business performance. The dataset contains **9,994 records and 11 columns** and has **no missing values or duplicate records**, indicating that the data is relatively clean for analysis.

The analysis shows that the business generated approximately **14.96 million in total sales** and **3.75 million in total profit**. Profit is moderately right-skewed, and the boxplot identifies some potential high-value outliers. These observations may represent genuine high-value transactions and should therefore be investigated rather than automatically removed.

A **moderate positive relationship exists between Sales and Profit**, with a correlation of approximately **0.61**. This indicates that higher sales generally tend to be associated with higher profit. In contrast, the scatterplot shows **no strong linear relationship between Discount and Profit**, suggesting that discount alone does not strongly explain changes in profitability.

From the category analysis, **Eggs, Meat & Fish** records the highest total sales, while **Oil & Masala** has comparatively lower sales. Regionally, the **West region** shows the strongest sales performance. However, the North region contains only one observation, so its performance should be interpreted cautiously.

The yearly trend analysis indicates that **sales increased from 2015 to 2018**, with **2018 recording the highest sales**. Overall, the EDA indicates positive business growth, a moderate association between sales and profit, differences in category and regional performance, and the presence of some potential profit outliers requiring further investigation.